MIL-C-3972B (PA)
Amendment 2
14 June 1974
SUPERSEDING
Amendment 1
16 May 1973

MILITARY SPECIFICATION

Cartridge, 60MM, Illuminating, M83A3
With Fuze, Time, M65A1
Loading, Assembling and Packing

This Amendment forms a part of Military Specification MIL-C-3972B (MU) dated 22 May 1972, and is approved for use by all departments and Agencies of the Department of Defense.

Paragraph 2.1 -Drawings.

Add the following alternate drawings:

"9278202 - Container, Ammunition, Fiber, PA 69: for 60MM Cartridge, Illuminating, M83A3 with Fuze, Time, M65A1, Packing and Marking.

9252725 - Container, Ammunition, METAL, PA 70: for 60MM Cartridge, Illuminating, M83A3 with Fuze, Time, M65A1.

9252726 - Box, Wirebound, Packing, Ammunition for 60MM Cartridge, Illuminating, M83A3 with Fuze, Time M65A1."

Add the following new paragraphs:

4.3.2.9 Unsealed Fiber Container (see dwg. 9278202).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Nose support missing.....	Visual	09001

FSC 1310

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Minor:	AQL 1.00 percent		
201.	Round can not be removed by hand	Manual	09002
202.	Evidence of poor workmanship	Visual	09003

4.3.2.10 Sealed Fiber Container (see dwg. 9278202).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Sealing strip incomplete or badly wrinkled	Visual	10001
Minor:	AQL 1.5 percent		
201.	Cover not properly seated	Visual	10002
202.	Contents loose	Manual	10003
203.	Marking misleading, unidentifiable, missing or incomplete	Visual	10004
204.	Evidence of poor workmanship	Visual	10005

4.3.2.11 Unsealed Metal Container (see dwg. 9252725).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Incorrect number of assemblies	Visual	11001
Minor:	AQL 1.00 percent		
201.	Spacers missing	Visual	11002
202.	Cushion(s) missing	Visual	11003

4.3.2.12 Metal Container (closed) (see dwg. 9252725).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Hardware improperly engaged	Visual	12001
102.	Container dented or deformed	Visual	12002
103.	Container punctured	Visual	12003
104.	Marking unidentifiable or missing	Visual	12004
Minor:	AQL 1.00 percent		
201.	Contents loose	Manual	12005
202.	Evidence of poor workmanship	Visual	12006

4.3.2.13 Box Wirebound Packing (see dwg. 9252726).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Wire missing, broken or loose	Visual	13001
102.	Board broken or split	Visual	13002
103.	Plastic bag missing	Visual	13003
Minor:	AQL 1.5 percent		
201.	Contents loose	Manual	13004
202.	Marking misleading, unidentifiable or missing	Visual	13005
203.	Car seal missing, unsealed or improperly positioned (when required)	Visual	13006
204.	Evidence of poor workmanship	Visual	13007

Paragraph 4.3.3 change Defect code 04017 to read:

TABLE II

Category	Defect	Code No.	ACC	60 Sample Size (see Note I) Rej	40 Sample Size (see Note I) Acc	40 Sample Size (see Note I) Rej
Major	Horizontal range less than 80% of horizontal range (see 3.3.5)	04017	2	3	2	3

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Add the following to paragraph 5.1.1:

"In addition, the open container (cover removed from body) shall be dried in a circulating oven for a minimum of 24 hours at a temperature of 120 F prior to packing. After temperature conditioning, the container shall be closed tightly for shipment, storage, etc. until no more than one-half hour prior to final packing of the cartridge in the container. The cartridge must be packed, and the fiber container sealed (Metal Packed) within 24 hours of the time the container was removed from the oven or from the moisture proof bag packed as described below. If the fiber containers are to be left on the assembly line or if for some reason the packaging in the fiber containers are to be stopped for more than one-half hour, the containers are to be closed tightly until they are ready to be used. Storage or transportation in excess of 24 hours after removal from conditioning oven shall require the cartons be packaged in moisture proof bags. Internal loading parts consisting of felt padding and chip board filler shall be included for moisture control by same oven drying requirements and subsequent packaging in moisture proof bags if necessary.

Add the following testing paragraph:

*4.3.3.4 Air leakage test-Major Defect-Code No. 08007- The Metal containers shall be tested for leakage with the procedure specified below and in accordance with the testing instructions contained in paragraph 4.4.3.

a. The Air leakage test shall be in effect 100% until 100 consecutive containers have been tested and the interior of the container visually inspected and found to be free of leakage.

b. Items tested which have passed the immersion test and found to be free of leakage by additional interior visual inspection shall be resealed and shall be released into production without further testing. Items which fail the immersion test or the visual interior inspection shall be rejected and removed from production.

c. Reduced inspection shall be instituted after satisfactorily completing (a) consisting of selecting a random metal container out of twenty-five (25) sealed metal containers consecutively produced.

d. If the container inspected in (c) above is found to be free of leakage as evidenced by immersion testing and visual interior

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inspection, the units represented by the sample shall be accepted without test. However, 100% visual exterior inspection evidence of improper sealing of the gasket is required on all units.

e. If the container inspected in (c) above is found to have leaked as evidenced by immersion testing or interior visual inspection, the defective container shall be rejected and the units represented by the sample shall also be rejected. The provisions of (a) shall then be instituted to requalify for sampling.

f. Whenever there is a lapse in production in excess of 30 (thirty) days or whenever there is a major change in manufacturing process requalification of reduced sampling shall be instituted as outlined in the paragraphs (a thru e) above.

Add the following new Test method:

*4.4.3 Air leakage test - The air leakage test shall be performed in accordance with paragraph 4.4.3 contained in specification MIL-P-116F. The sealed container immediately after sealing shall be immersed in a vessel of water (Temp. 50°-90°F) such that the top of the container is 3 ± 1 inch below the surface of the water. Evacuate the chamber containing the water to a vacuum of 3 ± 1/2 pound per square inch absolute. The assembly shall withstand vacuum submersion for 15-20 seconds without air leakage as defined in paragraph 4.4.3.6 of MIL-P-116F. The container shall be opened after the vacuum immersion test and shall be visually inspected for evidence of moisture or water.

Add the following new alternate paragraphs to "Preparation for Deliver":

5.1.2 Level A.-The cartridges shall be packed in accordance with dwg. 9278202 and dwg. 925725.

5.2 Packing.

5.2.2 Level A.-The packaged cartridges shall be packed in accordance with dwg. 9252726.

5.3 Marking.

5.3.2 The marking of the packages and pack shall be in accordance with dwg. 9278202, dwg. 9252725 and dwg. 9252726.

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*Asterisk indicates changes in addition to, or differing from, those of the previous Amendment.

Custodian:
Army - PA

Preparing Activity:
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